Practical-10

Aim: Mini Project: Analyze a real-world dataset, build an interactive dashboard, and present actionable insights with ethical considerations.

1. Introduction

Sales analysis helps businesses understand their sales performance, profit, customer demand, and product performance. It helps organizations identify profitable products, compare regions, and make better business decisions.

In this mini project, the Superstore Sales dataset is analyzed using Microsoft Power BI. The dashboard displays important business information through KPI cards, charts, tables, and interactive filters.

2 Objectives

- To analyze the Superstore Sales dataset.
- To study sales and profit across different regions.
- To compare product categories and sub-categories.
- To understand total orders and quantity sold.
- To identify high-performing products and regions.
- To create an interactive dashboard using Power BI.
- To present useful business insights and recommendations.

3 Tools and Technologies Used

Tool	Purpose
Google Colab	Data loading and preprocessing
Python and Pandas	Data cleaning and analysis
Excel/CSV	Dataset storage
Microsoft Power BI	Interactive dashboard creation

4 Dataset Description

The Superstore Sales dataset contains business sales information, including orders, products, categories, regions, sales, quantity, discount, and profit.

Column	Description
Order Date	Date when the order was placed
Product Name	Name of the product
Category	Product category
Sub-Category	Product sub-category
Region	Sales region

Column	Description
Sales	Total sales amount
Quantity	Number of units sold
Discount	Discount applied
Profit	Profit earned
Order ID	Unique order identification

5 Data Preprocessing

The Superstore Sales dataset was loaded and checked using Google Colab and Pandas. The following steps were performed:

1. Loaded the Excel dataset using Pandas.
2. Checked the first five rows and column names.
3. Checked missing values and duplicate records.
4. Corrected data types where required.
5. Prepared the cleaned dataset.
6. Imported the cleaned dataset into Microsoft Power BI.

6 Dashboard Design and Explanation

The dashboard is titled "Superstore Sales Dashboard".

6.1 KPI Cards

KPI Card	Description
Total Sales	Total sales amount
Total Profit	Total profit earned
Total Orders	Number of unique orders
Total Quantity	Total units sold

6.2 Charts and Visualizations

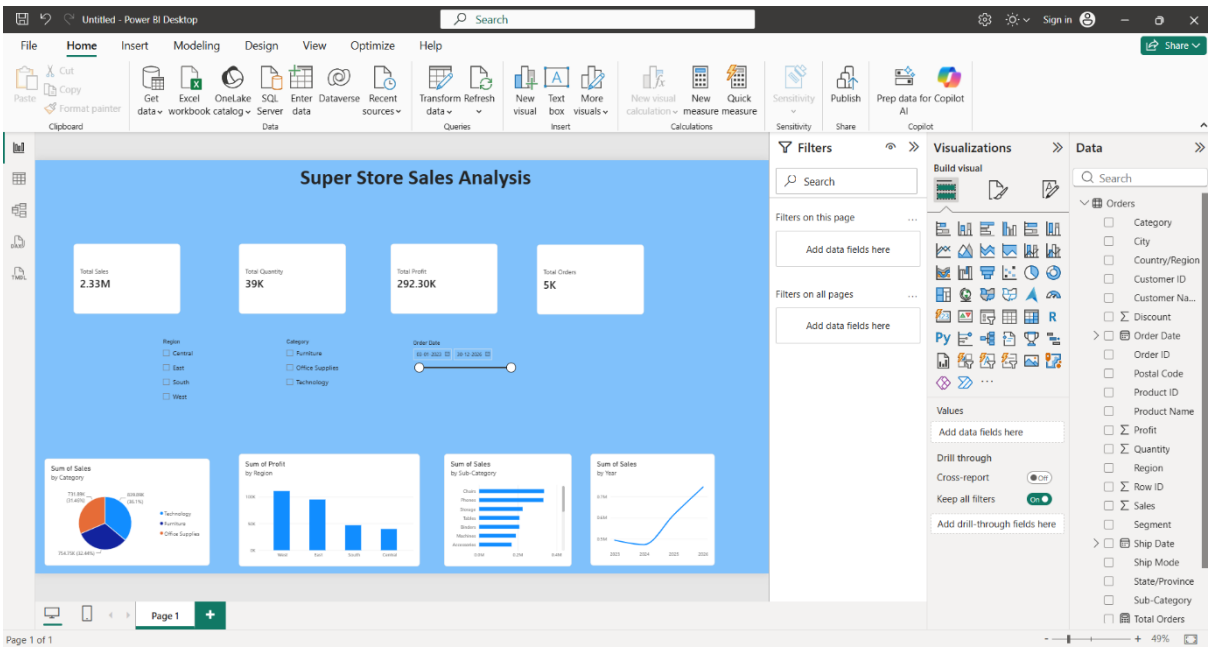
Chart	Purpose
Sales by Category	Compares sales across product categories
Profit by Region	Displays profit earned in each region
Sales by Sub-Category	Compares sales across sub-categories
Monthly Sales	Shows monthly sales trends
Sales and Profit Table	Displays detailed business information

6.3 Interactive Filters

- Region
- Category
- Order Date

Selecting a filter updates the dashboard visuals automatically.

7. Dashboard Screenshot



8 Analysis and Key Insights

The dashboard helps analyze the following business information:

- Total sales generated by the business.
- Total profit earned from sales.
- Total number of orders.
- Total quantity of products sold.
- Best-performing product categories.
- Profit comparison across different regions.
- Monthly sales trends.

9 Actionable Recommendations

1. Focus on product categories generating higher sales.
2. Improve sales strategies in regions with lower profits.
3. Monitor monthly sales trends to understand customer demand.
4. Review discounts to maintain better profitability.
5. Focus on products with good sales and profit performance.

6 Use dashboard filters to support business decisions.

10. Ethical Considerations

- Maintain the privacy of customer information.
- Keep business data secure.
- Present sales and profit values accurately.
- Avoid changing data to create misleading results.
- Use the dataset responsibly.
- Explain the limitations of the dataset.

11.Limitations

- Analysis depends on the quality of the provided dataset.
- Results are limited to the available sales records.
- The dashboard does not predict future sales.
- Charts show business patterns but do not prove the reasons behind them.
- Dashboard values change when filters are applied.

12. Result

The Superstore Sales dataset was successfully analyzed using Microsoft Power BI. An interactive dashboard was created with four KPI cards, charts, a table, and slicers. The dashboard helps understand sales, profit, orders, quantity, product categories, regions, and monthly sales performance.

13. Conclusion

This mini project provided practical experience in analyzing real-world sales data and creating an interactive dashboard using Microsoft Power BI. The dashboard makes business information easier to understand through KPI cards, charts, and filters. It helps identify sales trends, compare regions, and understand product performance. It also highlights the importance of accurate data analysis and responsible use of business information.